Development cooperation combines the concepts of development and cooperation and can be understood from a political perspective, where politics is closely related to power and operates in both formal and informal contexts. It exists at the intersection of technical practices and political processes.

Actors engage in development assistance for multiple reasons. These motivations may be **altruistic, economic, political, military, or humanitarian**. Development assistance is often used to pursue different objectives such as promoting economic growth, reducing poverty, improving governance, supporting infrastructure, ensuring loan repayment, combating terrorism, spreading specific ideologies, or advancing political and economic influence.

From an institutional perspective, development cooperation policy refers to a set of policies implemented by governments or multilateral institutions aimed at creating the conditions for sustainable economic and social development in other countries. These policies are carried out by governmental organizations or non-governmental organizations at national or international levels.

Development cooperation must be understood within a global context shaped by major challenges such as climate change, conflicts, international migration, and the emergence of new donors with different approaches. Other economic flows, such as foreign direct investment and remittances, often exceed development aid in importance. There is also an ongoing debate about the relevance of development cooperation, including critical perspectives on development models, alternative forms of development, post-development theory, and post-colonial approaches.

Recent global events, particularly the COVID-19 pandemic, have exposed structural vulnerabilities in health systems, environmental management, and socio-economic equality both within and between countries. The crisis highlighted the need for stronger public health systems, resilient supply chains, global partnerships, environmental protection, welfare systems, and poverty reduction. It also demonstrated the importance of international scientific cooperation, global alliances, and community participation.

Institutional transformations have also influenced development cooperation. At the international level, there has been a shift from the Millennium Development Goals to the Post-2015 Agenda and the Sustainable Development Goals (SDGs). At the national level, countries such as Italy have reformed their development cooperation systems, while universities increasingly contribute to development cooperation through research, training, and partnerships.

A key conceptual shift has occurred from “international development” to “global development.” Traditionally, international development focused mainly on improving conditions in poorer countries and addressing inequalities between developed and developing nations. In contrast, global development emphasizes the interconnected nature of globalized capitalism, global public goods such as health and environmental protection, the global challenge of sustainable development, the climate crisis, and inequalities both within and among countries. It also reflects the growing interdependence between the Global North and South.

The international development agenda evolved significantly with the adoption of the Millennium Declaration in 2000 and the subsequent adoption of the 2030 Agenda for

Sustainable Development in 2015 by the United Nations. The Sustainable Development Goals establish global objectives for social, economic, and environmental sustainability. They are structured around goals, targets, and indicators that measure progress. These indicators are periodically updated and provide measurable benchmarks to assess development outcomes.

The concept of sustainable development also builds on earlier work such as the Brundtland Commission's report *Our Common Future* (1983–1987), which emphasized the importance of balancing economic growth, environmental protection, and social wellbeing.

In the Italian context, development cooperation forms an integral part of foreign policy. Italian legislation, including Law 49 of 1987 and Law 125 of 2014, establishes development cooperation as a means to promote solidarity among peoples, protect human rights, reduce poverty and inequality, support sustainable development, prevent conflicts, and promote peace. Italian policy also emphasizes gender equality, humanitarian aid, and citizen participation in international solidarity.

According to international frameworks, development cooperation has several major tasks. These include supporting developing countries in guaranteeing universal social standards and human rights, promoting economic convergence and reducing global inequalities, and helping countries contribute to international public goods such as vaccines or fair trade systems.

To qualify as development cooperation, activities must meet four criteria. First, they must explicitly support national or international development priorities. Second, they must not be driven by profit, although they may encourage economic activities with positive developmental outcomes. Third, they must intentionally favor developing countries by addressing structural constraints that limit their development opportunities. Fourth, they must be based on cooperative relationships that enhance the ownership and autonomy of developing countries in shaping their own development strategies.

Development cooperation is therefore more than simply transferring financial resources. It includes actions such as reducing trade barriers for developing countries or supporting policy changes that improve development outcomes.

The main purposes of development cooperation include guaranteeing basic social protection standards, promoting convergence in living standards across countries, and supporting the provision of international public goods. Its main characteristics are that it supports development priorities, is not profit-driven, favors developing countries, and is based on cooperative partnerships. Development cooperation can take different forms, including financial or in-kind transfers, capacity building through training and technology cooperation, and policy change initiatives.

Various instruments and mechanisms are used in development cooperation, including budget support programs, solidarity taxes, syndicated loans, and social impact bonds, which link investment returns to measurable social outcomes. International partnerships and alliances, such as the GAVI, the Vaccine Alliance, demonstrate how global cooperation supports public health and development objectives.

It is important to distinguish development cooperation from related concepts. Development cooperation is broader than "aid," which typically refers only to concessional resource transfers from donor countries. It is also part of international

cooperation, although not all forms of international cooperation focus on development. The provision of international public goods, financing for development, and international public finance are broader categories that may include activities not specifically aimed at development or not focused on supporting developing countries.

Overall, development cooperation represents a complex field that combines political, economic, and social dimensions. It seeks to promote sustainable development, reduce global inequalities, strengthen international partnerships, and improve living conditions worldwide through cooperative and non-profit-oriented actions.

Development cooperation can be classified according to several dimensions. One classification concerns the type of resources transferred, such as financial resources, technology, or material support. Another classification is based on geographical destination, which includes North–South cooperation between developed and developing countries, South–South cooperation among developing countries, and trilateral cooperation involving multiple partners.

Development cooperation can also be categorized according to the actors involved and the channels through which cooperation occurs. These actors include governments, international organizations, non-governmental organizations (NGOs), and decentralized or local authorities. Actors can be distinguished by their nature, such as public or private, by their level of competence, such as local, regional, national, or international, and by their scale of action. Cooperation may take different forms, including bilateral cooperation between two countries, multilateral cooperation through international organizations, multi-bilateral cooperation combining different actors, and decentralized cooperation involving local authorities or NGOs.

In bilateral cooperation, central governments of donor countries typically work directly with central governments of recipient countries. Multilateral cooperation involves international organizations that manage resources contributed by donor governments. Multi-bilateral cooperation involves international organizations and donor governments working together, while decentralized cooperation includes local authorities, NGOs, and local recipients.

Another important dimension of development cooperation concerns resource flows and financing. Official Development Assistance (ODA) statistics, often collected by international institutions, help track financial flows and identify the composition of the donor community and the sources of development finance.

Development cooperation is also characterized by conditions attached to assistance. Conditions are requirements imposed by donors as part of lending or grant agreements. They can be formal and legally binding or informal expectations, and they may be imposed before assistance is granted (ex ante), after assistance is provided (ex post), or both. These conditions represent contractual terms that require recipients to undertake certain actions or achieve specific results in exchange for financial support.

Conditions may relate to eligibility requirements for obtaining aid, transparency and reporting procedures, use of funds, and achievement of specific results. They may focus on inputs such as expenditures, outputs such as service delivery results, outcomes such as policy effects, or broader impacts. Conditions can also concern macroeconomic

targets, structural reforms, governance and institutional changes, financial accountability, or public service delivery.

Donors require conditions for several reasons. They seek to ensure that loans will be repaid, promote recipient responsibility for development programs, maintain the integrity of funded operations, influence policy priorities, encourage reforms, and sometimes send political signals.

Development cooperation also varies according to intervention scenarios, which reflect different stages of crisis or development. In emergency situations, cooperation focuses on humanitarian relief, food aid, and the resettlement of displaced populations. In recovery and rehabilitation phases, interventions aim to rebuild infrastructure, restore productive sectors, and re-establish public services. In development scenarios, the focus shifts to implementing reforms, promoting participation, improving sectors of society, strengthening local ownership, and ensuring long-term sustainability.

Another way to classify development cooperation is by sector of intervention. These sectors include humanitarian assistance, climate change and natural resource management, conflict management, democracy and governance, basic education, economic development, global health, and water, sanitation, and hygiene. Although multiple sector classifications exist, development cooperation increasingly reflects the interconnected nature of development challenges, especially in relation to global sustainability goals.

Development cooperation can also be analyzed according to levels or scales of intervention. At the macro level, policies establish general rules or guidelines designed to address collective problems. Plans provide long-term strategies or roadmaps to achieve broad objectives, such as national development plans. Programs act as policy implementation tools by defining timing, budgets, and responsible actors. At the micro level, projects represent specific sets of tasks designed to achieve concrete outcomes within broader programs, such as training medical staff or distributing essential resources.

Overall, development cooperation can be defined as a wide and diverse set of actions carried out by multiple actors using different approaches and objectives to promote development for specific populations within defined times and spaces. It involves coordinated efforts across different sectors, scales, and contexts to support social, economic, and institutional development.

This lecture on development cooperation focuses on the role of traditional donors and the importance of the OECD Development Assistance Committee (DAC) in international development cooperation.

The Organisation for Economic Co-operation and Development Development Assistance Committee (DAC) is a unique international forum that brings together many of the world's largest providers of development aid. Its main objective is to promote development cooperation and related policies that contribute to sustainable development, including pro-poor economic growth, poverty reduction, and improvements in living standards in developing countries. The broader goal is to support a future in which countries no longer depend on aid.

A central concept defined by the DAC is Official Development Assistance (ODA). ODA refers to government aid that specifically promotes the economic development and welfare of developing countries. The DAC adopted ODA as the “gold standard” of foreign aid in 1969, and it remains one of the main sources of financing for development assistance.

To be classified as ODA, resource flows must meet specific criteria. They must be directed to countries and territories included on the DAC list of ODA recipients or to eligible multilateral development institutions. They must be provided by official agencies, including national or local governments or their executive bodies. In addition, the resources must be concessional in nature, meaning they include grants or soft loans with a significant grant component of at least 25 percent. The primary objective of these resources must be the promotion of economic development and welfare in developing countries.

Some forms of assistance are not considered ODA. For example, military aid, assistance primarily aimed at promoting the donor country’s security interests, or financial support with mainly commercial objectives such as export credits do not qualify as development assistance under DAC standards.

The OECD itself originated in 1960 with eighteen European countries together with the United States and Canada, and it has expanded over time. The DAC currently includes a group of member countries and organizations that provide development assistance, while other international financial institutions and organizations participate as observers or partners.

The DAC also defines which countries are eligible to receive ODA. The DAC list of ODA recipients is primarily used for statistical purposes to measure and classify aid flows from donor countries. It includes low- and middle-income countries defined by the World Bank based on gross national income per capita, except countries that are members of the G8 or the European Union. The list also includes all Least Developed Countries as defined by the United Nations.

In addition to identifying eligible recipient countries, the DAC maintains a list of ODA-eligible international organizations, including multilateral agencies, international non-governmental organizations, networks, and public–private partnerships. Contributions to these organizations may be reported as ODA when they support development objectives.

The OECD-DAC plays an important role in shaping global debates on development cooperation. It organizes high-level meetings that bring together ministers and international partners to discuss global development challenges and determine strategic responses. These meetings address issues such as climate change, biodiversity loss, food and water insecurity, and the effects of conflicts. They also emphasize the importance of climate resilience, sustainable development, poverty reduction, and international partnerships.

Another key function of the DAC is monitoring and evaluating development cooperation. It conducts peer reviews of member countries’ development policies and defines international standards and guidelines for development cooperation practices. It also collects and publishes statistical data on development finance, including resource flows and the composition of Official Development Assistance provided by donor countries.

Beyond ODA, new approaches have been developed to measure development finance more comprehensively. One of these is Total Official Support for Sustainable Development (TOSSD), which aims to increase transparency regarding the full range of financial resources supporting sustainable development. TOSSD includes not only traditional development assistance but also resources from South–South cooperation, triangular cooperation, multilateral institutions, and both emerging and traditional donors. This measure complements ODA by providing a broader picture of global financing for sustainable development and by monitoring new trends in development finance.

Overall, the OECD-DAC plays a central role in international development cooperation by defining standards, monitoring aid flows, identifying eligible recipients, promoting policy dialogue, and improving transparency in global development finance. Through these functions, it contributes to shaping the global system of development cooperation and supports efforts to promote sustainable development worldwide.

Module 2 of the Master ICT4D & Social Good at the University of Turin explores the relationship between digital technologies, global development, and social transformation. The module focuses on Information and Communication Technologies for Development (ICT4D), digital transition, and the challenges of building a fair, inclusive, and sustainable digital society. It examines how digital technologies influence international development processes, reshape global power structures, and create both opportunities and inequalities across different regions of the world.

A central goal of the module is to develop a critical understanding of the digital society within the context of international development. Students are encouraged to analyze global challenges associated with digital technologies, reflect on disparities between different parts of the world, and explore alternative approaches to development that are ethical, sustainable, and equitable. The course is guided by three main frameworks: the Sustainable Development Goals developed by the United Nations, Digital Humanism, and decolonial theory. Together, these frameworks provide tools for understanding the social impact of technology and for evaluating how digital transformation can support human well-being.

The module begins by examining the concept of the digital society and questioning how inclusive and fair it really is. It highlights that digital technologies shape social interactions, economic opportunities, and political systems, but also reproduce existing inequalities and power structures. Students are encouraged to reflect on global challenges such as access to digital technologies, digital governance, artificial intelligence regulation, data protection, social media regulation, and the protection of digital identities. The course raises questions about whether current global development frameworks adequately address the role of ICT and explores the possibility of designing specific development goals related to digital technologies.

A key topic in the module is the digital divide. The digital divide is presented not simply as unequal access to technology but as a complex set of social relationships reflecting broader inequalities. These include differences between urban and rural populations, wealthy and poor communities, educated and non-educated groups, and between the Global North and Global South. Through a decolonial perspective, the digital divide is understood as a structure of dominance that reflects historical patterns of colonialism,

economic inequality, and knowledge hierarchies. The digital society is therefore seen as both a space of innovation and a space where historical power relations continue to operate.

The course also discusses the concept of digital colonization and digital sovereignty. Digital colonization refers to situations in which countries or regions become dependent on external technological infrastructures, platforms, and services controlled by large technology companies or foreign powers. This concentration of digital power can lead to loss of control over data, reduced innovation capacity, and economic dependency. In response, the module emphasizes the importance of digital sovereignty, which involves promoting local innovation, open-source technologies, community ownership, and independence from dominant technology providers. Digital sovereignty aims to give users and communities greater control over their digital environments and technological futures.

Different models of development are examined throughout the module. Traditional international development approaches, dominant since the post–World War II period, focus on modernization through capital investment, industrialization, and technological deployment. These approaches often follow a linear model of innovation, where technology is developed and then transferred to developing regions. However, the module critiques this model for ignoring local contexts, cultural differences, and social realities. It also highlights the continued use of top-down approaches such as large-scale technological interventions that may fail to address local needs.

In contrast, the module introduces alternative models of ICT4D that emphasize human-centered and community-based approaches. These approaches involve participatory action research, bottom-up innovation, and collaborative design processes. Rather than imposing external solutions, researchers and developers work with communities to identify their needs, understand local contexts, and co-design technological solutions. This process involves context analysis, field visits, workshops, and iterative development, ensuring that technologies are adapted to the specific social, cultural, and economic environment in which they are implemented.

An important concept discussed is the “plug-in principle” for ICT4D design. According to this principle, technological interventions should improve existing systems rather than replace them, should be minimally disruptive, and should integrate indigenous practices with modern technologies. Researchers must understand local conditions, gain trust within communities, and develop solutions that can be replicated or adapted in other contexts. This approach emphasizes that successful technological development depends on social relationships, collaboration, and long-term engagement with local stakeholders.

The module includes several case studies that illustrate different approaches to ICT4D. Large-scale initiatives such as the One Laptop per Child project demonstrate the limitations of technology-driven solutions that do not fully consider local needs, infrastructure, and training requirements. In contrast, community-centered projects such as the eBario initiative in Malaysia show how local ownership, training, and sustainable infrastructure can lead to successful outcomes. The eBario project established community telecentres, supported digital education, and provided communication services while integrating technology with local cultural and social practices. Lessons from such projects emphasize the importance of community leadership, sustainable design, and technology as an enabler of broader social development rather than an end in itself.

The module also examines ICT4D initiatives in regions such as Sarawak, where digital technologies are used to promote rural connectivity, education, and economic development. These initiatives include building telecommunications infrastructure, establishing digital community centres, supporting digital entrepreneurship, and developing digital literacy programs. However, the case studies also reveal significant challenges, including infrastructure limitations, high implementation costs, low digital literacy, language barriers, and emerging cybersecurity risks. These examples demonstrate that digital development requires not only technological solutions but also social, economic, and institutional support.

Another important area explored is the use of ICT in low-resource environments. The course presents examples of technologies that operate without full internet access, such as radio-based communication systems, voice platforms, and SMS-based information services. Projects like the Tibaŋsim system combine radio, GSM networks, and voice technologies to share information within communities using locally available infrastructure. These solutions demonstrate how innovation can be adapted to resource constraints and highlight the importance of accessibility and contextual design.

The role of artificial intelligence in global development is also discussed. The module explores how AI can support inclusive development by enabling local language processing, improving access to information, and supporting community-based data collection. However, challenges such as data availability, infrastructure limitations, algorithmic bias, and ethical concerns must be addressed. Participatory approaches to AI development, including stakeholder workshops and community engagement, are emphasized as ways to ensure that AI technologies benefit local populations.

The module further examines different theories and models of innovation. The traditional linear model of innovation describes technological progress as a sequence of invention, adoption, and diffusion. This model is criticized for presenting technology as deterministic and universal. Alternative models, such as ecosystem models and complex adaptive systems, emphasize the interaction between technology and society, highlighting feedback loops between social and technical factors. The course argues that innovation is shaped by political, economic, and cultural forces rather than being purely technical or neutral.

A critical perspective on innovation is developed by challenging common myths, such as the belief that technology is value-neutral, that innovation is purely technological, or that regulation hinders progress. Instead, the module shows that technological development is embedded in social values, power structures, and economic interests. The concentration of power in large technology companies, the influence of digital platforms on governance and public discourse, and the role of algorithms in shaping behavior illustrate the political nature of technology.

Ethics and social responsibility are central themes of the course. Scientists and technologists are presented as having a responsibility to consider the societal impact of their work and to act in the public interest. The module discusses how ICT has evolved from technical tools for communication into technologies that shape governance, influence behavior, and restructure democratic processes. It also critiques reductionist views of human beings as computational systems and highlights the importance of inclusive perspectives in technological development.

Finally, the module emphasizes the importance of democratizing technology and innovation. It explores whether technological development can be made more participatory and inclusive, allowing diverse communities to contribute to shaping digital futures. The course suggests that building a fair digital society requires collaboration, public debate, education, policy engagement, and awareness of global challenges. Rather than viewing technological progress as inevitable, the module encourages active participation in shaping the direction of digital transformation.

Overall, Module 2 presents a comprehensive and critical examination of ICT for development, highlighting the complex relationship between technology, society, and global inequality. It promotes a vision of digital transformation that is human-centered, ethically grounded, and socially responsible, emphasizing community participation, local innovation, and digital sovereignty as essential components of a more equitable and sustainable digital society.

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1. ICT4D and Digital Society

- **ICT4D (Information and Communication Technologies for Development)**
  - Studies how digital technologies support global development.
  - Focus on social, economic, and political impacts of technology.
  - Technology creates opportunities but also inequalities.
- **Digital society**
  - Digital technologies shape:
    - social interactions
    - economic opportunities
    - political systems
    - governance structures
  - Key questions:
    - Is digital society inclusive?
    - Is it fair and sustainable?
    - Who benefits from digital transformation?
- **Global digital challenges**
  - digital governance
  - data protection and privacy
  - AI regulation
  - social media regulation
  - digital identity protection

2. Key Theoretical Frameworks

- **Sustainable Development Goals (SDGs)** — United Nations
  - Human well-being and sustainable global development.
  - Technology as a tool for development.
- **Digital Humanism**
  - Human-centered technology.
  - Ethical and socially responsible innovation.

- **Decolonial theory**
  - Critiques Eurocentric views of development.
  - Highlights global power inequalities.
  - Focus on knowledge diversity and local perspectives.

3. Digital Divide

- Not only lack of access → **structural inequality**.
- Exists between:
  - urban vs rural areas
  - rich vs poor populations
  - educated vs non-educated groups
  - Global North vs Global South
- Causes:
  - economic inequality
  - historical colonial structures
  - knowledge hierarchies
  - infrastructure differences
- Digital divide reflects broader power relations.

4. Digital Colonization and Digital Sovereignty

- **Digital colonization**
  - Dependence on foreign technology platforms.
  - Control by large tech companies.
  - Loss of data control and innovation capacity.
- **Digital sovereignty**
  - Local control of technology.
  - Open-source solutions.
  - Community ownership.
  - Local innovation and autonomy.

5. Models of Development

Traditional Development Model

- Post–World War II modernization approach.
- Focus:
  - industrialization
  - capital investment
  - large-scale technological deployment
- Characteristics:
  - top-down
  - technology transfer from developed countries
  - ignores local context → often ineffective.

Alternative / Human-Centered ICT4D

- Bottom-up development.
- Community participation.
- Context-sensitive design.

- Participatory action research.
- Co-design with local communities.

6. ICT4D Design Principles

- **Participatory approach**
  - Field visits.
  - Needs assessment.
  - Community collaboration.
- **Plug-in principle**
  - Improve existing systems.
  - Minimal disruption.
  - Integrate local knowledge.
  - Technology adapts to society (not vice versa).
- **Requirements for success**
  - trust building
  - cultural awareness
  - long-term engagement
  - scalability and sustainability

7. Case Studies in ICT4D

- **Large-scale initiatives**
  - Example: One Laptop per Child.
  - Limitations:
    - lack of infrastructure
    - insufficient training
    - poor adaptation to local needs.
- **Community-based initiatives**
  - Example: eBario (Malaysia).
  - Key elements:
    - local ownership
    - telecentres
    - digital education
    - sustainable infrastructure.
- **Rural development initiatives**
  - digital community centres
  - digital entrepreneurship
  - connectivity projects
  - digital literacy programs.
- **Common challenges**
  - high costs
  - infrastructure limits
  - language barriers
  - cybersecurity risks.

8. ICT in Low-Resource Environments

- Technology adapted to limited infrastructure:

    - radio systems
    - SMS-based services
    - voice platforms
    - GSM networks.
- Goal:
    - accessibility
    - local usability
    - cost efficiency.

9. Artificial Intelligence for Development

- Potential benefits:
    - local language technologies
    - improved access to information
    - community data collection.
- Challenges:
    - data scarcity
    - algorithmic bias
    - infrastructure limits
    - ethical issues.
- Emphasis on:
    - participatory AI design
    - stakeholder involvement.

10. Innovation Models

- **Linear innovation model**
    - invention → adoption → diffusion.
    - criticized as deterministic.
- **Alternative models**
    - innovation ecosystems.
    - complex adaptive systems.
    - interaction between social and technical factors.
- Innovation shaped by:
    - politics
    - culture
    - economy
    - power relations.

11. Critical Perspective on Technology

- Technology is **not neutral**.
- Innovation reflects:
    - social values
    - economic interests
    - power structures.
- Issues:
    - concentration of power in tech companies
    - influence of algorithms

    - impact on democracy.

12. Ethics and Responsibility

- Scientists and technologists have social responsibility.
- Technology affects:
    - governance
    - public behavior
    - democratic processes.
- Need for:
    - ethical design
    - public interest focus
    - inclusive innovation.

13. Democratizing Technology

- Make innovation participatory.
- Encourage public debate and collaboration.
- Promote inclusive digital futures.
- Technology should empower communities.

The module on **Research and Fieldwork in International Cooperation** explains the role of research in development practice and prepares students to design their Final Project within the ICT4D (Information and Communication Technologies for Development) framework. The course focuses on applied research for social good, emphasizing the relationship between research, fieldwork, and processes of social change in international cooperation.

The main aim of the module is to help students understand what research means in development contexts, why research is important in international cooperation, how to design a research proposal, and how to develop their Final Project. Research in this Master is not a traditional academic thesis, not a descriptive internship report, and not purely theoretical work. Instead, it is **applied research for development**, meaning research-in-practice that emerges from field experience, integrates theory and practice, and aims to produce practical outcomes that generate social impact.

The Final Project has a dual structure. First, it includes a light theoretical component that provides a conceptual framework explaining the problem, justifying the chosen approach, and connecting ICT and development. This is not a long literature review but a focused theoretical basis. Second, it includes a practical component such as a prototype, a methodological framework, an adaptation of an ICT tool, or another output usable by an organization. The project is change-oriented, embedded in international cooperation, and focused on social good, even when the impact is small or local.

Research in international cooperation starts from real-world problems rather than theoretical questions. The field shapes the research question, and researchers collect feasible data instead of ideal data. Observations are transformed into structured knowledge, and ICT, theory, and practice are connected in meaningful ways. The Final Project represents the formalization of this process.

Research is closely linked to development because development work aims to produce social change. Research contributes to understanding people, contexts, and problems, and supports development policy and practice throughout all stages of a project. It is used in baseline studies, feasibility studies, pilot projects, ethnographic research, monitoring, and evaluation. Through research, development actors can gain information, better understand needs, and design appropriate interventions.

Research is essential to correctly understand problems and identify effective solutions. For example, in a Kurdish community in Northern Iraq, low school attendance was initially attributed to poverty, but research revealed that children were avoiding school due to violence and abuse by teachers. This led to more appropriate interventions such as teacher training. Another example shows that lack of contextual knowledge can produce harmful development programs, such as building schools for nomadic communities in Somaliland, which created security risks and failed to consider their mobile lifestyle. These examples demonstrate that research ensures development programs respond to real needs and avoid negative consequences.

The module explains two main research approaches: quantitative and qualitative research. Quantitative research focuses on numbers, measurement, and statistical analysis, asking how many people share certain characteristics or experiences. It uses structured methods and standardized data collection. Qualitative research focuses on meanings, experiences, and social processes, exploring what people think, feel, and why. It uses open and flexible methods adapted to context. In practice, development research often combines both approaches.

A quantitative study of children in a refugee camp might measure health indicators, educational outcomes, or behavioural statistics using standard measures. A qualitative approach would instead focus on the perspectives of children and families, observing their experiences and asking open questions about their situation. The choice of research approach depends on the research goals, desired social change, and researchers' beliefs about how change occurs.

Developing a research project involves several main steps: selecting a research problem, defining research questions, choosing research methods, collecting data, analysing data, and presenting findings. However, the process is not strictly linear and often moves back and forth between stages.

The research process typically begins with curiosity and field experience. Researchers identify a topic, define what they want to know and why, formulate a research question, choose a methodology, select methods for data collection, determine where and with whom research will take place, address ethical issues, analyse data, and communicate results. Ethical considerations include managing power relations, ensuring informed consent, protecting confidentiality, and minimizing risks for participants.

An example of research design in ICT4D involves studying the use of a mobile data collection platform in a rural development project implemented by a local NGO. Field observations may reveal problems such as inconsistent data quality, incomplete use of the platform, or continued reliance on paper-based practices. These observations become a research problem because they affect decision-making and project effectiveness. A research question might examine how field staff use and adapt the digital tool and how this affects project implementation. The research may use qualitative methods such as interviews, observation, and document analysis, focusing on user practices and contextual

factors. The results can lead to practical recommendations, such as improved workflows, staff training, or adaptations to the ICT system.

The module emphasizes the importance of literature review in ICT4D research. Literature review helps situate field experience within existing knowledge, interpret observations, refine research questions, justify research choices, and build a conceptual framework. In international cooperation, literature review often follows field observation rather than preceding it. Researchers first encounter a problem in practice and then consult literature to better understand it. Literature does not replace fieldwork but interacts with it, serving as a lens for interpretation.

Relevant literature in ICT4D includes studies on digital inclusion, technology appropriation, information systems in low-resource contexts, participatory design, and critical development studies. A literature review should identify key debates, compare perspectives, and connect concepts to field observations. It should be concise and focused rather than a summary of all available research.

The course also provides strategies for locating relevant literature, such as consulting recent reviews, searching across different disciplines, using academic databases, identifying key terms, scanning texts efficiently, and maintaining focus on the research question. Effective reading involves recording key information, scanning content, focusing on relevant sections, and extracting central ideas.

A key stage in research design is defining research questions. These questions must be feasible, researchable, and answerable within time and resource constraints. Researchers often begin with a long list of possible questions and then narrow them down based on feasibility, access to data, available skills, and ethical considerations. A focused research question allows deeper analysis and stronger conclusions, while trying to answer too many questions leads to superficial results.

Good research questions clearly define the research goal, the phenomenon studied, the context, the participants, and the nature of the inquiry. They specify what is being studied, where, and why. For example, a research question might explore how field staff adapt a digital tool, what challenges they encounter, and how organizational factors influence technology use.

The module also highlights the importance of planning research activities through a research timetable. A timetable outlines research phases, activities, and timing, ensuring that the project is realistic and feasible. Typical phases include initial observations, literature review, research design, data collection, data analysis, writing, and final presentation.

Overall, the module presents research as a practical and context-driven process that connects field experience, theoretical understanding, and social change. Research in international cooperation is not only about generating knowledge but also about improving development practice, designing effective interventions, and producing socially beneficial outcomes. Fieldwork raises questions, literature helps interpret them, and research connects theory and practice to support meaningful change in development contexts.